

Sujith S

Student Researcher | Machine Learning • Natural Language Processing • AI Systems • Data Science
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Research Summary

Full-time Computational Statistics and Data Analytics student at VIT University (2022-2027), returning to the program after internship. Published researcher with a peer-reviewed conference paper on machine learning and statistical modeling applied to financial time-series data. Experienced building large-scale AI systems across natural language processing, deep learning, multi-agent reasoning, retrieval-augmented generation, and distributed data engineering using Python, Java, and JavaScript; committed to advancing the state of the art in intelligent systems through collaborative, impact-driven research.

Education

Int.Msc Computational Statistics and Data Analytics

VIT University | 2022-2027

Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Statistical Modeling & Inference, Algorithmic Foundations of Optimization, Data Structures & Algorithms, Probability & Statistics, Software Engineering, Computer Networks

Harvard CS50 Programs: [CS50AI](#) (search, optimization, ML, neural networks), [CS50x](#), [CS50P](#), [CS50SQL](#), [CS50R](#)

Research & Publications

- **Stock Market Analysis and Prediction using Machine Learning** – [El Ruha International Conference on Scientific Research](#), 14th Proceedings, pp. 730-740, 2025. Investigated time-series forecasting and statistical modeling for financial market prediction; designed ML pipelines for feature engineering, model training, and real-world equity data evaluation. [\[Code\]](#) [\[Proceedings\]](#)

Skills

Programming: Python, Java, JavaScript, TypeScript, SQL, R

ML & AI Research: Deep Learning, NLP, Natural Language Understanding, LLMs, RAG, Multi-Agent Systems, Reinforcement Learning, Recommendation Systems, Sentiment Analysis, Statistical Inference, Time-Series Forecasting, Algorithmic Optimization, Probabilistic Modeling

Frameworks & Data Science: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, LangChain, LangGraph, CrewAI, AutoGen, XGBoost, FAISS, ChromaDB, Apache Kafka, Apache Spark, MLflow, Pandas, NumPy

Engineering & Tools: FastAPI, PostgreSQL, Redis, Docker, Git, Linux, CI/CD, AWS, GCP, Distributed Systems, REST APIs

Research Projects

Autonomous Research Agent

[GitHub](#)

Python, LangChain, LangGraph, RAG, FAISS, ChromaDB, Hugging Face Transformers, Sentence Transformers, FastAPI, PostgreSQL, Redis

- Designed and implemented a fully autonomous research pipeline using Retrieval-Augmented Generation, LangGraph multi-step reasoning, and semantic vector search (FAISS, ChromaDB) to decompose complex queries, retrieve and validate sources, and synthesize structured research reports; built NLP preprocessing using Hugging Face Transformers for dense embedding generation and similarity-ranked knowledge retrieval over large unstructured corpora.

Multi-Agent Business Automation System

[GitHub](#)

Python, FastAPI, CrewAI, LangGraph, LangChain, AutoGen, PostgreSQL, Redis, Docker

- Formulated and built a multi-agent AI system using CrewAI, LangGraph, and AutoGen where specialized autonomous agents collaborate via contextual memory and intelligent task routing; researched and implemented supervisor-agent orchestration, inter-agent communication protocols, and distributed workflow execution supporting real-time sentiment analysis and predictive analytics at scale.

Social Media Sentiment Lakehouse

[GitHub](#)

Python, Apache Kafka, Apache Spark Streaming, Delta Lake, Hugging Face Transformers, NLTK, MLflow, PostgreSQL, Redis

- Built a large-scale real-time NLP analytics platform using Kafka, Spark Streaming, and Hugging Face Transformers for sentiment classification, emotion detection, and topic modeling; engineered a Bronze-Silver-Gold Delta Lake architecture with ACID compliance and integrated MLflow for experiment tracking and automated model retraining over evolving data distributions.

Work Experience

Founder

[Shoot the Breeze](#) | 2023 - Present

- Investigated, designed, and deployed AI-powered intelligent systems integrating LLMs, multi-agent architectures, and NLP pipelines using Python and FastAPI, leading full research-to-deployment cycles from problem formulation and architecture design through implementation, evaluation, and production deployment.

App/Web Developer

[CropSky](#) | Pre-incubated at VIT-TBI Research & Innovation Hub | 2024 - Present

- Developed backend systems and REST APIs within a university-affiliated research incubator at VIT-TBI, contributing data-driven software solutions to real-world agricultural challenges and collaborating with cross-functional teams to align application development with scientific research objectives.

Certifications & Achievements

- [Google Generative AI Leader](#)
- [Google Cloud Computing Foundations](#)
- [FreeCodeCamp Legacy Full Stack](#)
- [OCI 2025 Certified Developer Professional](#)
- AWS [Solutions Architect](#) and [Developer](#)
- [Software Engineer Certification - HackerRank](#)